SEW EURODRIVE INSTALLATION, MAINTENANCE, AND LUBRICATION INFORMATION EXTRACTED FROM SEW EURODRIVE OPERATION AND MAINTENANCE MANUAL FOR GEAR UNIT SERIES

Technical Note

Mounting Procedures – Footed Reducers

Mechanical

Drives must be properly installed if they are to produce rated torque. Incorrectly installed drives may experience oil leaks, reduced life spans, or even catastrophic failure. An appropriate mounting site should contain the following:

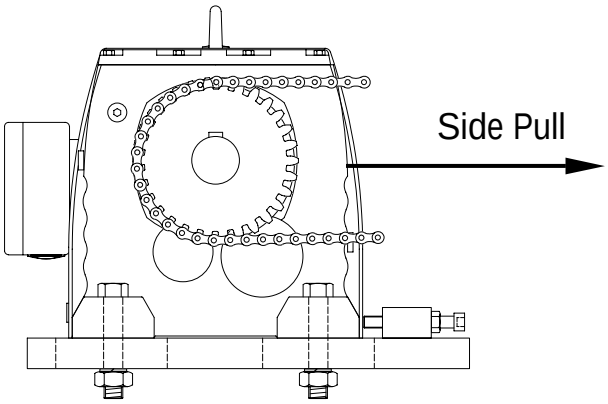
- Ambient temperatures between 0°C and 40°C (32°F to 104°F).
- Unimpeded airflow to and around the units.
- Accessibility to oil drain, level and breather plugs.
- Adequate space for removing the fan guard from the brake motor when adjusting or replacing the brake.
- A mounting surface that is flat, torsionally rigid, and dampened against vibration. Flatness tolerances should not exceed the following values:

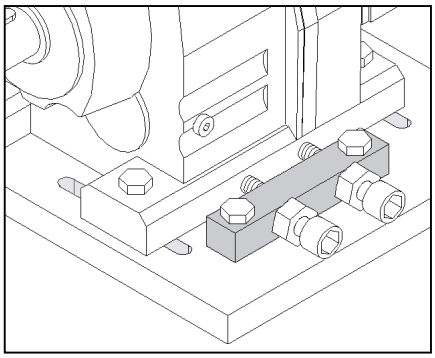
  Reducer Sizes 87 and smaller: 0.004 inches
  Reducer Sizes 97 and larger: 0.008 inches

1. Solid Shaft

SEW reducers have oversized mounting holes to allow for minor adjustments in alignment. Therefore, the largest bolt that can fit through the hole is not necessarily the best choice. SEW lists recommended bolt sizes in the reducer sections of the SEW Tech Manual.

Foot-mounted units that are subject to side pulls caused by chains or belts should include dowel pins or tightening bolts in addition to foot mounting bolts. As shown in the right picture below, a metal bar with two tightening bolts is anchored to a mounting plate either by a weld or by bolts that thread into the plate. This design prevents the reducer from creeping towards the load and from acquiring excessive slack in a belt or chain if the bolts in the reducer feet loosen. Plus, it still allows the use of slots in the mounting plate, which provides flexibility when initially positioning the reducer.

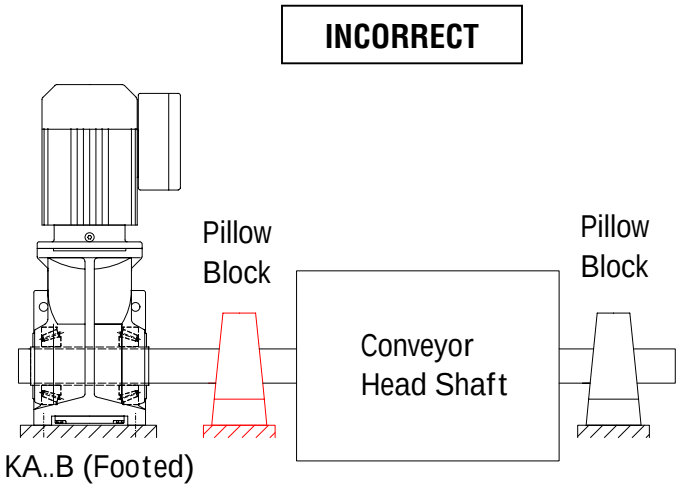




2. Hollow Shaft

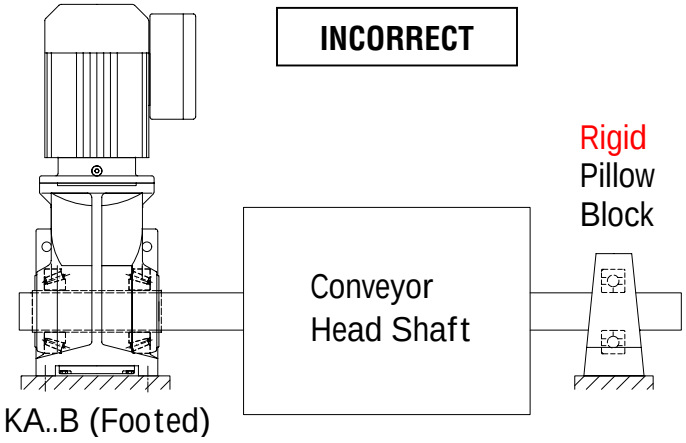
A gear reducer with a hollow shaft and feet is an acceptable means of support for one end of a machine's shaft. However, special care must be exercised during mounting to assure that the reducer is not subject to a bending moment (preload) caused by misalignment of the bearings in the system. Therefore, SEW strongly recommends the use of a self-aligning bearing on the opposite end of the machine's shaft.

Listed below are correct and incorrect methods to mount a footed unit with a hollow shaft.



Problem: There are four bearings that must be critically aligned. The closer the pillow block bearing is to the reducer, the more critical is the alignment.

Solution: Remove the pillow block bearing between the reducer and the head pulley, allowing the reducer to serve as support bearings for the head pulley.

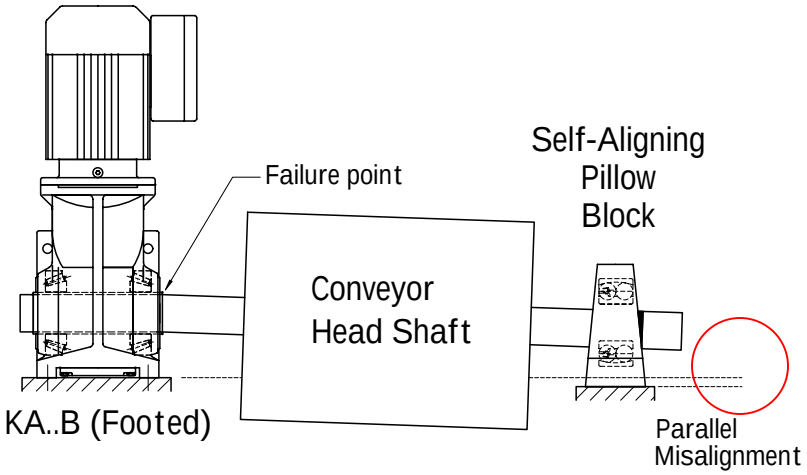


Problem: Due to its inflexibility, the rigid pillow block bearing can cause preloading on reducer bearings if the conveyor shaft is not correctly aligned.

Solution: Replace the rigid pillow block bearing with a self-aligning type.

Technical Note

INCORRECT

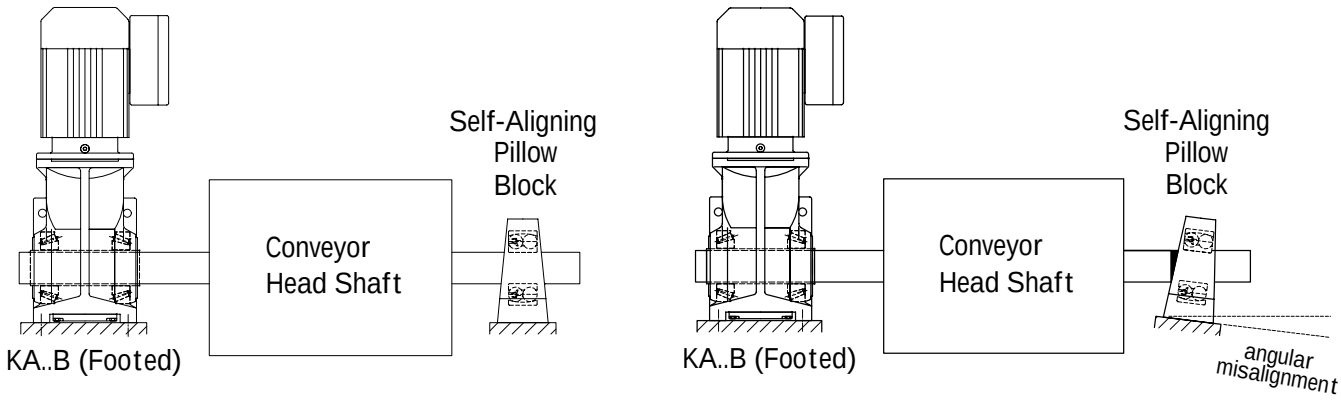


Problem: Even though a self-aligning bearing is used, the parallel offset caused by different mounting planes creates a bent conveyor shaft and a bending moment (i.e. an overhung load) at the edge of the reducer. The result of this overhung load is either premature failure of the reducer bearings or fatigue failure of the conveyor shaft.

Solution: Ensure that the mounting surfaces are on the same plane to keep conveyor shaft straight. Use shims where necessary.

CORRECT

Allow the gear reducer to support one end of the conveyor shaft and a self-aligning bearing to support the other end of the conveyor shaft. A self-aligning bearing tolerates some angular misalignment from an unleveled surface, as shown in the right diagram. Since the conveyor shaft remains straight, an overhung load is avoided. Notice the difference between a parallel misalignment (above) and an angular misalignment (below).



Technical Note

Mounting Procedures – Installing Shaft Components



While installing components on a solid shaft, it is important to follow correct mounting procedures to avoid unnecessary damage. Hammering on the shaft can create Brinell marks (indentations) on the bearing raceway, causing premature bearing failure.

All of the SEW 7-series reducers are supplied with a tapped hole on the output shaft as an aid when installing couplings, sheaves, or sprockets. The following table shows the size and depth of the threads.

Inch:

Shaft Diameter	Thread Size	Thread Depth (in)
0 to 13/16”	¼ - 20	0.630
7/8” to 15/16”	5/16 – 18	0.870
1.0” to 1-1/8”	3/8 – 16	0.870
1-1/4” to 1-3/8”	½ - 13	1.120
1-1/2” to 1-7/8”	5/8 – 11	1.380
2.0” to 3-1/4”	¾ - 10	1.610
3-3/8” to 4-3/4”	1 – 8	2.130

Metric:

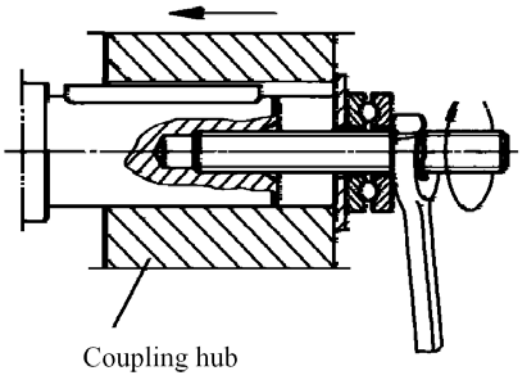
Shaft Diameter	Thread Size	Thread Depth (mm)
16 mm to 21 mm	M6	16
22 mm to 24 mm	M8	19
25 mm to 30 mm	M10	22
31 mm to 38 mm	M12	28
39 mm to 50 mm	M16	36
51 mm to 85 mm	M20	42
86 mm to 130 mm	M24	50
Over 130 mm	M30	63

Technical Note

Figure 1 shows the correct procedure for mounting couplings or hubs onto an output shaft. In some cases, the thrust bearing at the mounting jig is not required.

Tightening the nut pushes the coupling hub onto the output shaft.

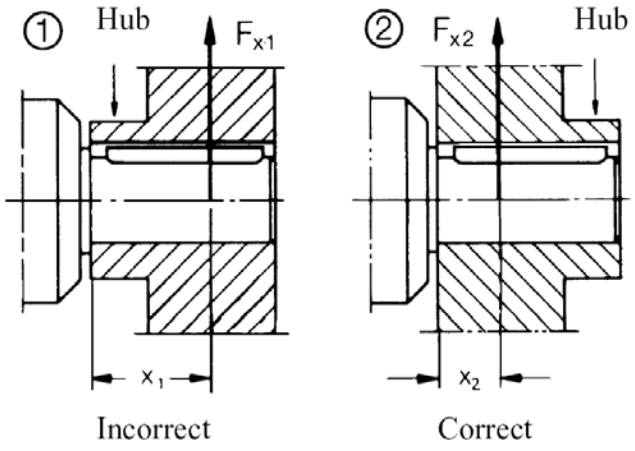
Figure 1 – Installation



The arrangement of the protruding hub should also be carefully observed. The load force, F_x, should be as close to the reducer housing as possible to maximize the overhung load capacity of the reducer.

Figure 2 shows correct and incorrect arrangements of a sprocket or gear hub. Notice how the hub is correctly placed away from the reducer housing in order to minimize the distance, x, from the reducer.

Figure 2 - Arrangement



For easier installation, apply a lubricant to the bore of the hub, or preheat the component to approximately 175°F.

Technical Note

Mounting Procedures – Bolt Sizes and Torques

This technical note details the proper method for choosing and tightening the mounting bolts for SEW gear reducers, flanges, and torque arms.

Reducers with Feet

Foot-mounted reducers fasten in place by bolts slipped through the mounting holes in the mounting feet and secured with a nut beneath the mounting surface. To accommodate misalignment, SEW slightly oversizes mounting holes. Bolts should be long enough to protrude past the fastening nut a distance equal to the diameter of the bolt – refer to Figure 1. For additional information about proper mounting procedures for footed units, refer to SEW Technical Note **GM-019**.

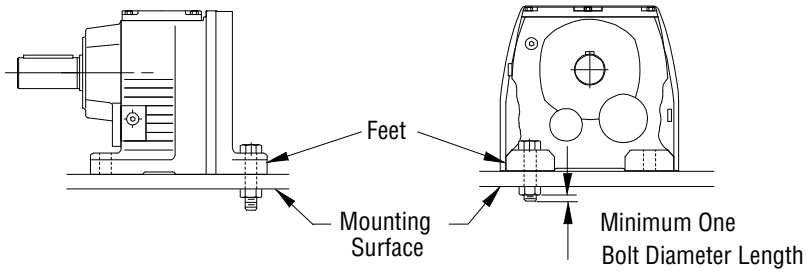


Figure 1: Minimum Bolt Length

Gear Unit				SAE (Inch)		ISO (mm)	
R	K	S	W	Hole Ø	Bolt Ø	Hole Ø	Bolt Ø
R07			W10, W20	0.26	1/4	6.6	M6
R17, R27, R37		S37	W30	0.35	5/16	9	M8
RX57	K37, K47	S47, S57		0.43	3/8	11	M10
RX67, R47, R57	K57, K67	S67		0.53	7/16	13.5	M12
R67				0.55	7/16	14	M12
R/RX77, R/RX87	K77	S77		0.69	5/8	17.5	M16
R/RX97, RX107	K87	S87		0.87	3/4	22	M20
R107	K97	S97		1.02	7/8	26	M24
R137	K107, K167			1.30	1-1/4	33	M30
R147, R167	K127, K157, K187			1.54	1-3/8	39	M36

Table 1: Mounting Hole and Bolt Diameters

NOTE: Minimum bolt property class is 8.8 (ISO) or grade 8 (SAE).

Reducers with Rail Holes (F Series)

Unlike foot-mounted reducers with through-holes, SEW F-series reducers have a rail mount consisting of tapped mounting holes on the sides of the reducer, as shown in Figure 2.

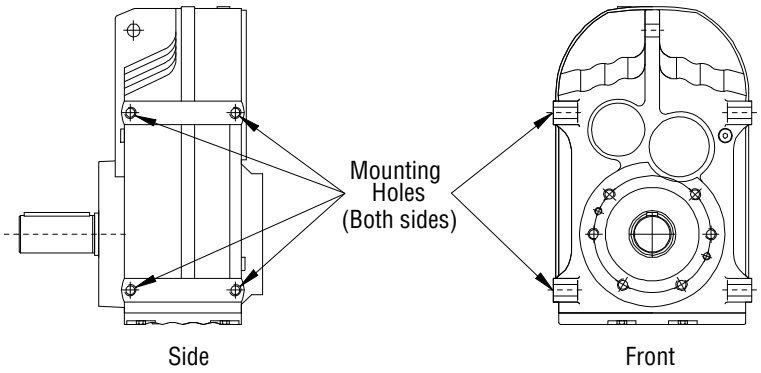


Figure 2: F-Series Mounting Holes

Table 2 shows the tap depths of F-Series reducers. The maximum bolt length to ensure that the bolt does not bottom out inside the mounting hole is equal to the thickness of the mounting surface through which the bolt must pass plus a distance that is slightly less than the tap depth. The minimum bolt length to ensure sufficient thread engagement is equal to the thickness of the mounting surface plus a distance equal to the bolt diameter. **Exception**: The F27 requires a minimum bolt engagement equal to 1.5 times the bolt diameter, due to its cast aluminum construction.

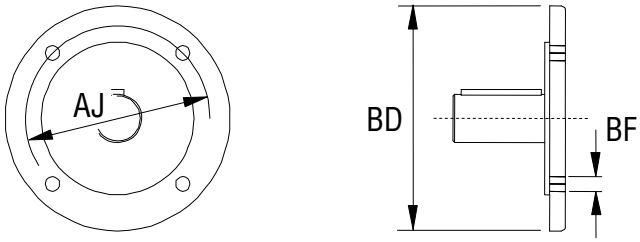
Gear Unit	Bolt Diameter	Tap depth	
		in	mm
F27*	M8	0.63	16
F37	M8	0.43	11
F47	M10	0.59	15
F57, F67	M12	0.67	17
F77, F87	M16	1.02	26
F97	M20	1.10	28
F107	M24	1.42	36
F127	M30	1.77	45
F157	M36	2.17	55

Table 2: F-Series Mounting Holes

NOTE: Minimum bolt property class is 8.8 (ISO) or grade 8 (SAE).

Reducers with Flange

Table 3 shows the recommended bolt diameters for each flange size. For more information about proper mounting procedures for flanged reducers, refer to SEW Technical Note **GM-020**.



Flange Size (BD Dimension)		Flange Dimensions				Bolt Diameter	
		in		mm			
in	mm	AJ	BF	AJ	BF	in	mm
3.15	80	2.56	0.26	65	6.6	1/4	6
4.33	110	3.43	0.35	87	9	5/16	8
4.72	120	3.94	0.26	100	6.6	1/4	6
5.51	140	4.53	0.35	115	9	5/16	8
6.30	160	5.12	0.35	130	9	5/16	8
6.30	160	5.12	0.33	130	8.5	5/16	8
7.87	200	6.50	0.43	165	11	3/8	10
9.84	250	8.46	0.53	215	13.5	7/16	12
11.81	300	10.43	0.53	265	13.5	7/16	12
13.78	350	11.81	0.69	300	17.5	5/8	16
17.72	450	15.75	0.69	400	17.5	5/8	16
21.65	550	19.69	0.69	500	17.5	5/8	16
25.98	660	23.62	0.87	600	22	3/4	20

= FF27 only

Table 3: Flange Bolt Diameters

NOTE: Minimum bolt property is class 8.8 (ISO) or grade 8 (SAE).

EXCEPTION: Bolts should be class 10.9 (ISO), grade 10 (SAE), or better for the following reducers: RF37F with 120mm flange; R47F with 140mm flange, R57F with 160mm flange.

Reducers with Torque Arms

Table 4 shows the recommended bolt sizes to mount an SEW torque arm onto a reducer. The torque arm mounts to the bottom of a K-series reducer and mounts to the side of an S or W-series reducer. For information regarding the proper design of torque arms brackets, refer to SEW Technical Note **GM-021.**

Gear Unit	Bolts			Torque
	Qty.	Dia. (mm)	Length (mm)	Nm
KA37	4	M10	25	48
KA47	4	M10	30	48
KA57	4	M12	35	86
KA67	4	M12	35	86
KA77	4	M12	40	86
KA87	4	M16	40	210
KA97	4	M20	50	410
KA107	4	M24	60	710
KA127	4	M36	130	2500
KA157	4	M36	130	2500
SA37	4	M6	16	11
SA47	4	M8	25	25
SA57	4	M8	25	25
SA67	4	M12	35	86
SA77	4	M12	35	86
SA87	4	M16	45	210
SA97	4	M16	50	210
WA10	4	M6	12	11
WA20	4	M6	12	11
WA30	4	M6	16	11

Table 4: Torque Arm Bolt Diameters

NOTE: All bolts should be of property class 8.8 (ISO).

Bolt Torque Requirements

Table 3 shows the torque limits for SAE and ISO bolts.

Bolt Size		Class ISO 8.8 and SAE 8 Bolts[1]				Class ISO 10.9 and SAE 10 Bolts[1]			
		lb-in		Nm		lb-in		Nm	
		T[2]	+10%	T[2]	+10%	T[2]	+10%	T[2]	+10%
M4	1/8	27	29	3	3.3	43	47	4.8	5.3
M5	3/16	53	58	6	6.6	84	93	9.5	10.5
M6	1/4	100	106	11	12	140	154	16	18
M8	5/16	240	270	27	30	350	390	40	44
M10	3/8	480	530	54	60	710	780	80	90
M12	7/16	820	900	93	100	1240	1360	140	150
M14	1/2	1310	1440	148	160	1950	2150	220	240
M16	5/8	2040	2240	230	250	2990	3290	338	370
M18	11/16	2910	3200	329	360	4150	4570	469	520
M20	3/4	4110	4510	464	510	5850	6440	661	730
M24	7/8	7060	7770	798	880	10050	11010	1136	1250
M27	1-1/8	10410	11450	1176	1300	14820	16300	1674	1840
M30	1-1/4	14130	15550	1597	1760	20130	22140	2274	2500
M36	1-3/8	24590	27040	2778	3060	35020	38520	3957	4350

Table 3: Bolt Tightening Torques

NOTE: [1] SAE bolt torques are based on standard UNC thread for that nominal bolt diameter; [2] T = tightening torque.

CAUTION: The +10% values indicate the upper torque limits; do not tighten bolts beyond this limit.

6 Inspection/maintenance



⚠ WARNING

Risk of injury if the drive starts up unintentionally.

Severe or fatal injuries.

- Disconnect the drive from the power supply before you start working on the unit.
- Prevent the drive from starting up unintentionally for example, by locking the key switch or removing the fuses from the current supply, and attach a warning sign that prohibits switching on the drive.



⚠ WARNING

Risk of injury if preloaded shaft connections are loosened.

Severe or fatal injuries.

- Before releasing any shaft connections, make sure there is no active torsional torque present that could lead to tension within the system.



⚠ WARNING

Risk of burns due to hot gear unit and hot gear unit lubricant.

Severe injuries.

- Let the gear unit cool down before you start working on it.
- Carefully remove the oil level plug and the oil drain plug.

NOTICE

Loss of lubricant qualities due to filling of wrong lubricant.

Damage to the gear unit.

- Do not mix synthetic lubricants and mineral lubricants.
- Do not mix different synthetic lubricants.

NOTICE

Damage to oil seal caused by cleaning the gear unit with a high pressure device.

Gear unit damage.

- Do not clean the gear unit with a high-pressure cleaning device.

NOTICE

Damage to gear unit due to ingress of foreign objects during maintenance and inspection work.

Gear unit failure.

- Prevent foreign particles from entering into the gear unit during maintenance and inspection work.

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INFORMATION

Maintain the inspection and maintenance intervals. This is necessary to ensure operational safety.

INFORMATION

Perform a safety check and functional check following maintenance and repair work.

6.1 Wearing parts

Gearing

If you observe the SEW-EURODRIVE design criteria and the inspection and maintenance intervals, then the gearing components of the gear units will be wear-free after the run-in period. The worm gearing is an exception for design reasons. Depending on the operating conditions, material on the tooth flanks of the worm gear is removed to different extents. The main influencing factors are:

- Speed
- Load
- Operating temperature
- Lubricant (type, viscosity, additives, pollution)
- Switching frequency

For information on the worm gearing service life under certain operating conditions, contact SEW-EURODRIVE.

Rolling bearings

Rolling bearings in the gear unit, adapter, and input shaft assembly have a limited service life, even under ideal operating conditions. This nominal bearing service life is a solely statistical value. The actual service life of an individual bearing may deviate greatly from this value. The main influencing factors are:

- Speed
- Equivalent bearing load
- Operating temperature
- Lubricant (type, viscosity, additives, pollution)
- Lubricant supply of the bearing
- Misalignment under operating load

The rolling bearings must therefore be inspected regularly. Observe the corresponding inspection and maintenance intervals in the chapters "Inspection/maintenance intervals" (→ 🗎 136), "Lubricant change intervals" (→ 🗎 137), "Maintaining AL../AMS../AM../AQS../AQ../EWH.. adapters" (→ 🗎 138) and "AD.. input shaft assembly maintenance" (→ 🗎 138).

For information on the nominal bearing service life under certain operating conditions, contact SEW-EURODRIVE.

Lubricants

Lubricants are subject to aging. Their service life is limited depending on the load conditions.

The service life depends significantly on the oil operating temperature. The dependency of lubricant change intervals on the operating temperature is depicted in the figure in chapter "Lubricant change intervals" (→ 🗎 137).

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Oil seals

Oil seals are contact seals that seal unit housings at emerging elements, such as shafts, from the environment. Oil seals are wear parts with a service life that is influenced by the following factors, among others:

- Shaft speed and circumferential speed at the sealing lip
- Ambient conditions (temperature, dust, humidity, pressure, chemicals, radiation)
- Lubricant (type, viscosity, additives, pollution)
- Surface quality of the sealing
- Lubricant supply of the sealing
- Oil seal material

Due to the various influencing factors, it is not possible to predict the service life. Therefore the oil seals must be inspected regularly. Observe the corresponding inspection and maintenance intervals in the chapters "Inspection/maintenance intervals" (→ 🗎 136), "Lubricant change intervals" (→ 🗎 137), "Maintaining AL../AMS../AM../AQS../AQ../EWH.. adapters" (→ 🗎 138) and "AD.. input shaft assembly maintenance" (→ 🗎 138).

Coupling ring

The couplings installed in the AMS.., AM.., AL.., AQS.., AQ.., and EWH.. adapters are designed to be positive, puncture-proof and low-maintenance claw couplings with an impact and vibration-absorbing cam ring (AMS.., AM.., EWH..) or coupling ring (AQS..., AQ.., AL..). The service life is determined by the following factors, among others:

- Ambient conditions (temperature, chemicals, radiation)
- Operational conditions (switching frequency, impact characteristics)

Adhere to the corresponding inspection and maintenance intervals in chapter "Maintaining AL../AMS../AM../AQS../AQ../EWH.. adapters" (→ 🗎 138).

Rubber buffer

The rubber buffer is required for shaft-mounted gear units of the F and W gear unit types for torque support. Rubber buffers are wear parts with a service life that is influenced by the following factors:

- Load
- Ambient conditions
  - Temperature
  - Humidity
  - Aggressive chemicals, e.g. ozone
- Switching frequency
- Impact characteristics

Flexible bushing

A so-called flexible bushing is required for the torque arm of the S and K gear unit types. Flexible bushings are wear parts with a service life that is influenced by the following factors:

- Load
- Ambient conditions
  - Temperature
  - Humidity
  - Aggressive chemicals, e.g. ozone

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- Switching frequency
- Impact characteristics

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6.2 Inspection/maintenance intervals

The following table lists the obligatory intervals and the corresponding measures:

Time interval	What to do?
• Every 3000 operating hours, at least every 6 months	• Check oil and oil level • Check running noise for possible bearing damage • Visual inspection of the seals for leakage • Check that all screw plugs, any oil sight glass, the breather valve, and the gear unit cover screws are tight. • For gear units with a torque arm: Check and replace the rubber buffers, if necessary
With mineral oil: • Depending on the operating conditions (see illustration in chapter "Lubricant change intervals" (→ 🗎 137)), every 3 years at the latest • According to oil temperature	• Change the oil • Replace rolling bearing grease (recommendation) • Replace oil seal (do not install it in the same track again)
With synthetic oil: • Depending on the operating conditions (see illustration in chapter "Lubricant change intervals" (→ 🗎 137)), every 5 years at the latest • According to oil temperature	• Change the oil • Replace rolling bearing grease (recommendation) • Replace oil seal (do not install it in the same track again)
• Varying (depending on external factors)	• Touch up or renew the surface/ anti-corrosion coating • Check operation of breather valve (if present)
• From 5th year of operation	• Check the blocking effect of the backstop annually. Ensure that the maximum blocking torque is not exceeded.

Exceptions

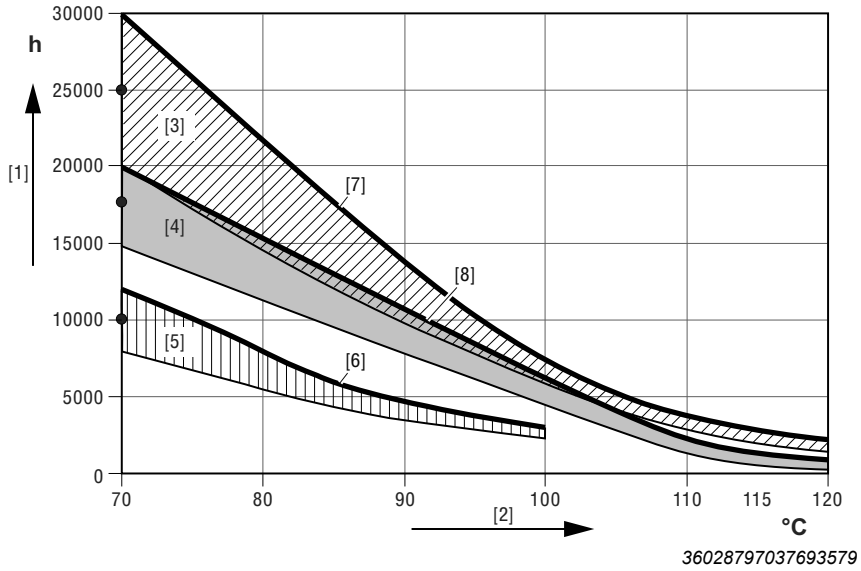
The following gear units are lubricated for life. A scheduled oil change is not necessary:

- R..07, R..17, R..27 helical gear units
- F..27 parallel-shaft helical gear unit
- SPIROPLAN® gear units

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6.3 Lubricant change intervals

Use the following figure to determine the number of operating hours between 2 oil changes based on the sustained oil bath temperature at normal ambient conditions. In case of special designs under severe/aggressive ambient conditions, change the lubricant more frequently.



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[1] Operating hours
[2] Continuous oil bath temperature
[3] CLP PG/CLP PG NSF H1
[4] CLP HC/ CLP HC NSF H1
[5] CLP (CC)/E
[6] GearOil Base by SEW-EURODRIVE
[7] GearOil Poly (H1) / GearFluid Poly (H1) by SEW-EURODRIVE
[8] GearOil Synth (H1) by SEW-EURODRIVE

- Average value for each oil type at 70 °C

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6.4 Maintaining AL../AMS../AM../AQS../AQ../EWH.. adapters

The following table shows the time intervals to be observed and the corresponding measures:

Time interval	What should I do?
• Every 3000 operating hours, at least every 6 months	• Check the running noises to detect possible bearing damage. • Visually check the test adapter for leakage. • With the drain hole design, check whether the condensation drain holes are clear.
• After 10 000 operating hours	• Check the rotational clearance. • Visually check the coupling ring (AMS.., AM.., EWH.. or AQS.., AQ.., AL..).
• After 10 000 operating hours with NBR/FKM oil seals • After 20 000 operating hours with Premium Sine Seal adapter oil seals	• Change the oil seal. With standard NBR or FKM oil seals, the new oil seal must not be fitted on the previous track. This is allowed with Premium Sine Seal adapter oil seals.

6.5 AD.. input shaft assembly maintenance

The following table lists the obligatory intervals and the corresponding measures:

Time interval	What to do?
• Every 3000 operating hours, at least every 6 months	• Check the running noises to detect possible bearing damage. • Visually check the adapter for leakage.
• After 10 000 operating hours	• Change the oil seal. Do not mount it in the same track.

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6.6 Inspection/maintenance of the gear unit

6.6.1 Checking the oil level and changing the oil

The procedure when checking the oil level and changing the oil depends on gear unit type, size and mounting position. Determine the code letter (A, B, C, D or E) in the following table in regard of gear unit type and size. Use the code letter to find the reference for the procedure for the corresponding gear unit in the 2nd table.

Gear unit type	Size	Code letter for chapter "Checking the oil level and changing the oil"					
		M1	M2	M3	M4	M5	M6
R	R..07 to 27	B					
	R..37/R..67	A					
	R..47/R..57	A				B	A
	R..77 to 167	A					
	RX..57 to 107	A					
F	F..27	B					
	F..37 to 157	A					
K	K..19/K..29	C					
	K..39/K..49	A					
	K..37 to 187	A					
S	S..37	C					
	S..47 to 97	A					
W	W..10 to 30	B					
	W..19 to 59	B					
	W..37 to 47	D			E	D	

Code letter	Chapter "Checking the oil level and changing the oil"	Reference
A:	• Helical gear units • Parallel-shaft helical gear units • Helical-bevel gear units K..39/K..49, K..37 to 187 • Helical-worm gear units S..47 to 97 **With oil level plug**	(→ 🗎 140)
B:	• Helical gear units • Parallel-shaft helical gear units • SPIROPLAN® W..0/W..9 **Without oil level plug, with cover plate**	(→ 🗎 143)
C:	• S..37 helical-worm gear units • K..19/K..29 helical-bevel gear units **Without oil level plug, without cover plate**	(→ 🗎 147)

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Code letter	Chapter "Checking the oil level and changing the oil"	Reference
D:	• SPIROPLAN® W..37/W..47 **In mounting positions M1, M2, M3, M5, M6 with oil level plug**	(→ 🗎 150)
E:	• SPIROPLAN® W..37/W..47... **In mounting position M4 without oil level plug and cover plate**	(→ 🗎 152)

Refer to chapter "Mounting positions" (→ 🗎 155) for notes on the mounting positions.

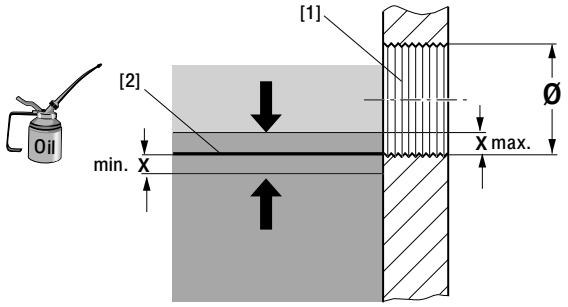
You cannot check the oil level of gear units in pivoted mounting position. The gear units are delivered with the correct oil level. Observe the specifications and fill quantities on the nameplate if you have to change the oil.

6.6.2 A: Helical, parallel-shaft helical, helical-bevel and helical-worm gear units with oil level plug

Checking the oil level at the oil level plug

Proceed as follows to check the oil level of the gear unit:

1. Observe the information at the beginning of chapter "Inspection/maintenance" (→ 🗎 132).
2. Determine the positions of the oil level plug and the breather valve using the mounting position sheets. See chapter "Mounting positions" (→ 🗎 155).
3. Place a container underneath the oil level plug.
4. Slowly unscrew the oil level plug. Small amounts of oil may leak out as the permitted maximum oil level is higher than the lower edge of the oil level bore.
5. Check the oil level according to the following figure and the corresponding table.



634361867

[1] Oil level bore
[2] Oil level setpoint
X Min./max. oil level

Ø oil level bore	Approved fluctuation "x" of the oil level mm
M10 × 1	1.5
M12 × 1.5	2
M22 × 1.5	3
M33 × 2	4
M42 × 2	5

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6. Proceed as follows if the oil level is too low:
   - Remove the breather valve from the breather bore.
   - Fill in fresh oil of the same type (contact SEW-EURODRIVE if necessary) via the breather bore, up to the lower edge of the oil level bore.
   - Screw in the breather valve again. When doing this, observe the tightening torques in chapter "Tightening torques for oil level plugs, oil drain plugs, screw plugs, breather valves and oil sight glasses" (→ 🗎 36).
7. Screw in the oil level plug again. When doing this, observe the tightening torques in chapter "Tightening torques for oil level plugs, oil drain plugs, screw plugs, breather valves and oil sight glasses" (→ 🗎 36).

Checking the oil via the oil drain plug

Proceed as follows to check the gear unit oil:

1. Observe the information at the beginning of chapter "Inspection/maintenance" (→ 🗎 132).
2. Determine the position of the oil drain plug using the mounting position sheets. See chapter "Mounting positions" (→ 🗎 155).
3. Remove a little oil from the oil drain plug.
4. Check the oil consistency:
   - Viscosity (have this carried out by a suitable laboratory if necessary)
   - If you can see that the oil is heavily contaminated, it is advisable to change the oil, even if this is outside the service intervals specified in "Inspection and maintenance intervals" (→ 🗎 136).
5. Check the oil level. See previous chapter.

Changing the oil via the oil drain plug and the breather valve



⚠ WARNING

Risk of burns due to hot gear unit and hot gear unit oil.

Severe injuries.

- Let the gear unit cool down before you start working on it. Due to the better flowability, the gear unit oil should still be warm so that the gear unit can be drained best.

1. Observe the information at the beginning of chapter "Inspection/maintenance" (→ 🗎 132).
2. Determine the position of the oil drain plug, the oil level plug and the breather valve using the mounting position sheets. See chapter "Mounting positions" (→ 🗎 155).
3. Place a container underneath the oil drain plug.
4. Remove the oil level plug, the breather valve and the oil drain plug.
5. Drain all the oil.
6. Re-insert the oil drain plug. When doing this, observe the tightening torques in chapter "Tightening torques for oil level plugs, oil drain plugs, screw plugs, breather valves and oil sight glasses" (→ 🗎 36).

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7. Fill in fresh oil of the same type (contact SEW-EURODRIVE if necessary) via the breather bore. Do not mix different synthetic lubricants.
   - Observe the oil quantities according to the specifications on the nameplate or according to the mounting position. See chapter "Lubricant fill quantities" (→ 🗎 215).
   - Check the oil level at the oil level plug.
8. Re-insert the oil level plug and the breather valve. When doing this, observe the tightening torques in chapter "Tightening torques for oil level plugs, oil drain plugs, screw plugs, breather valves and oil sight glasses" (→ 🗎 36).

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8.2 Lubricants

Unless a special arrangement is made, SEW-EURODRIVE supplies the drives with a lubricant fill adapted for the specific gear unit and mounting position. The mounting position (see chapter "Mounting positions" (→ 🗎 155)) must therefore be specified in the drive order. You must adapt the lubricant fill in case of any subsequent changes made to the mounting position (see chapter "Lubricant fill quantities" (→ 🗎 215)).

8.2.1 Bearing greases

The gear unit rolling bearings are given a factory-fill with the greases listed below. SEW-EURODRIVE recommends re-greasing the rolling bearings with a grease filling at the same time as changing the oil.

The table shows the lubricants recommended by SEW-EURODRIVE:

Area of application	Ambient temperature	Manufacturer	Type
Standard	-40 °C to +80 °C	SEW-EURODRIVE	Grease HL 2 E1[1)]
		Fuchs	Renolit CX-TOM 15[1)]
		Klüber	Petamo GHY 133 N
 2)	-40 °C to +40 °C	SEW-EURODRIVE	Grease HL 2 H1 E1
		Bremer & Leguil	Cassida Grease GTS 2
 3)	-20 °C to +40 °C	Fuchs	Plantogel 2S

1) Bearing grease based on semi-synthetic base oil.
2) Lubricant for the food processing industry.
3) Easily biodegradable lubricant for environmentally sensitive areas.

The following grease quantities are required:

- **For fast-running bearings (gear unit input side):** Fill the cavities between the rolling elements one-third full with grease.
- **For slow-running bearings (gear unit output side):** Fill the cavities between the rolling elements two-thirds full with grease.

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8.2.2 Lubricant table

The lubricant recommendation provided in the lubricant table in no way represents a guarantee as to the quality of the lubricant delivered by each respective supplier. Each lubricant manufacturer is responsible for the quality of their product.

Oils of the same viscosity class from different manufacturers do not have the same characteristics. In particular, the minimum and maximum permitted oil bath temperatures are manufacturer-specific. These temperatures are specified in the lubricant tables.

The oil viscosity and type (mineral/synthetic) to be used are determined by SEW-EURODRIVE specifically for each order. This information is noted in the order confirmation and on the gear unit's nameplate. If you use other lubricants for the gear units and/or use the lubricants at temperatures outside the recommended temperature range, SEW-EURODRIVE does not assume liability.

The values specified in the lubricant tables apply as of the time of printing of this document. The data of the lubricants are subject to dynamic change on the part of the lubricant manufacturers. For the latest information about the lubricants, visit: **www.sew-eurodrive.de/lubricants**.

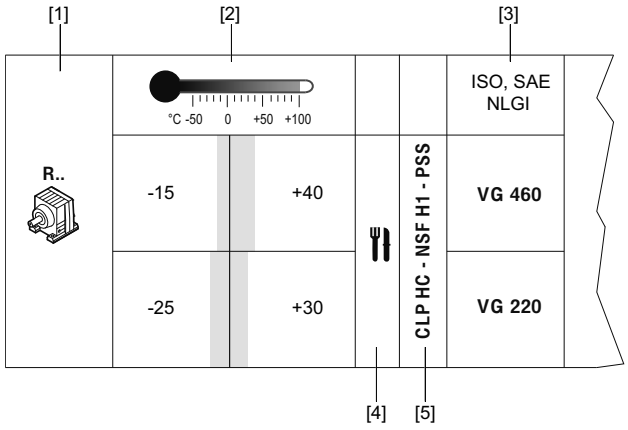
NOTICE

Damage to the gear unit due to improper lubricants.

Possible damage to property.

- Do not mix synthetic lubricants.
- Do not mix synthetic lubricants and mineral lubricants.

Information on the table structure



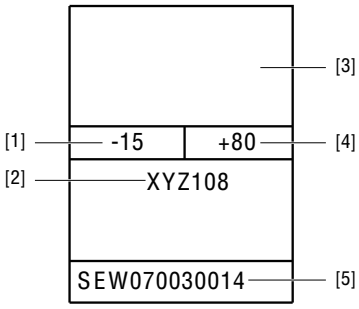
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[1] Gear unit type
[2] Ambient temperature range
[3] Viscosity class
[4] Note on special approvals
[5] Lubricant type

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The specified ambient temperatures are guide values for selecting a suitable lubricant. The exact upper and lower temperature limits for project planning are specified in the table with the respective trade name. Bear in mind during project planning that the viscosity increases at low temperatures and that this might influence the starting behavior.

Information on the various lubricants



[1] Lowest oil sump temperature in °C, **going below this value during operation is not permitted**
[2] Trade name
[3] Manufacturer
[4] Highest oil sump temperature in °C. The service life will be considerably reduced when this temperature is exceeded. Observe the lubricant change intervals according to chapter "Lubricant change intervals" (→ 🗎 137).
[5] Approvals regarding compatibility of the lubricant with approved oil seals

Lubricant compatibility with oil seals

Approval	Explanation
SEW07004_ _13:	A lubricant especially recommended with regard to compatibility with the approved oil seals. The lubricant exceeds the state-of-the-art requirements regarding elastomer compatibility.

Approved application temperature range of the oil seals

In the low temperature range, oil seals can withstand shaft deflections (e. g. through overhung load) only to a limited extent. Especially avoid or limit pulsating or changing radial displacements of the shaft. Contact SEW-EURODRIVE, if required.

Oil seal Material class	Permitted Oil sump temperature
NBR	-40 °C to +80 °C
FKM	-25 °C to +115 °C
FKM-PSS	-25 °C to +115 °C

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Limitations of use of oil seals with the specific lubricant are described in the following table:

Material class			Manufacturer		Material	
S	1	NBR	1	Freudenberg		72 NBR 902
			2	Trelleborg		4NV11
	2	FKM	1	Freudenberg	1	75 FKM 585
					2	75 FKM 170055
			2	Trelleborg	1	VCBVR
			3	SKF	1	FKM 00934

Examples:

S11: Only the elastomer 72NBR902 from Freudenberg meets the requirements of the approval in conjunction with the specific lubricant.

S2: Only the elastomer FKM meets the requirements of the approval in conjunction with the specific lubricant.

Key

The following table shows the symbols and abbreviations used in the used in the lubricant table and their meaning:

Abbreviation/symbol	Meaning
	Synthetic lubricant (marked gray)
	Mineral lubricant
CLP	Mineral oil
CLP PG	Polyglycol (PG)
rPCF	reduced Product Carbon Footprint
CLP HC	Synthetic hydrocarbons to polyalphaolefins (PAO)
E	Ester-based oil
	Lubricant for the food processing industry and feed industry. Oils are NSF-H1 registered and compliant in accordance with FDA 21 CFR § 178.3570
	Lubricants with particularly reduced CO_2 footprint (cradle-to-gate) with sustainable raw materials.
	Lubricants with slight bio-degradability for environmentally sensitive areas (> 60% according to OECD 301 or according to appendix A of EPA 2013 VGP)
	Lubricant suitable for explosion-protected gear units and gearmotors
1)	Helical-worm gear units with CLP-PG: Contact SEW-EURODRIVE.
2)	Low-viscosity grease
3)	With appropriate measures, the gear units can be operated at ambient temperatures as low as -40 °C. Contact SEW-EURODRIVE.

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Abbreviation/symbol	Meaning
Oil seal	Oil seal
Premium Sine Seal	Oil seal of the Premium Sine Seal type. The addendum "PSS" for the lubricant type indicates compatibility with the sealing system.

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Lubricant table for R.., F.., and K..7 gear units

The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).

R.. RES K..7 KES HK.. F..	[3] °C -50 0 +50 +100	[1]	[2]	ISO,SAE NLGI	SEW EURODRIVE	bremer & leguil	Castrol	FUCHS	Mobil	KLÜBER LUBRICATION	Shell	SINOPEC	TOTAL
[4]	-15 +40		CLP	VG 220	-15 +80 GearOil Base 220 E1/US1/CN1/BR1 SEW07004313		-15 +80 Optigear BM 220	-15 +80 Renolin CLP 220 Plus SEW07004313	-15 +80 Mobilgear 600 XP 220 SEW07004013	-15 +80 Klüberoil GEM 1-220 N	-15 +80 Shell Omala SG 220	-15 +80 AP-SGO 220 SEW07004313	-15 +80 Cater EP 220
	-20 +30			VG 150	-20 +80 GearOil Base 150 E1/US1/CN1/BR1 SEW07004313		-20 +80 Optigear BM 150	-20 +80 Renolin CLP150 Plus SEW07004313	-20 +80 Mobilgear 600 XP 150 SEW07004013	-20 +80 Klüberoil GEM 1-150 N	-20 +80 Shell Omala SG 150	-20 +80 AP-SGO 150 SEW07004313	-20 +80 Cater EP 150
[4]	-15 +40		CLP PSS	VG 220	-15 +80 GearOil Base 220 E1/US1/CN1/BR1 SEW07004313			-15 +80 Renolin CLP 220 Plus SEW07004313	-15 +80 Mobilgear 600 XP 220 SEW07004013			-15 +80 AP-SGO 220 SEW07004313	
	-20 +30			VG 150	-20 +80 GearOil Base 150 E1/US1/CN1/BR1 SEW07004313			-20 +80 Renolin CLP150 Plus SEW07004313	-20 +80 Mobilgear 600 XP 150 SEW07004013			-20 +80 AP-SGO 150 SEW07004313	

[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🖹 200).

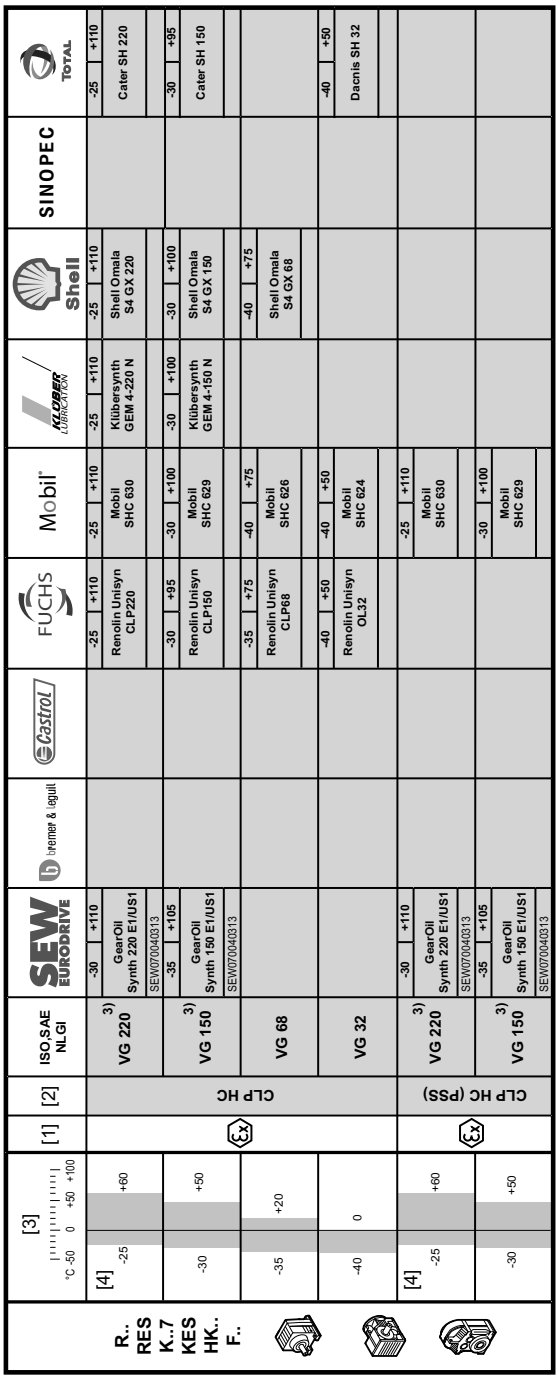
	[3] °C -50 0 +50 +100	[1]	[2]	ISO, SAE NLGI	SEW EURODRIVE	Lubrigard & Legault	Castrol	FUCHS	Mobil	KLÜBER LUBRICATION	Shell	SINOPEC	TOTAL
R.. RES K..7 KES HK.. F..	[4] -25 +80		CLP PG	VG 220	-25 | +115 GearOil Poly 220 E1 SEW07004031 3		-25 | +115 Optigear Synthetic 800/220	-25 | +115 Renolin PG220	-25 | +115 Mobil Glycoyle 220	-25 | +115 Klübersynth GH 6-220	-25 | +115 Shell Omala S4 WE 220		-25 | +115 Cater SY 220
	-30 +70			VG 150	-30 | +115 GearOil Poly 150 E1 SEW07004031 3					-30 | +115 Klübersynth GH 6-150			
	[4] -25 +80		CLP PG (PSS)	VG 220	-25 | +115 GearOil Poly 220 E1 SEW07004031 3					-25 | +115 Klübersynth GH 6-220			
	-30 +70			VG 150	-30 | +115 GearOil Poly 150 E1 SEW07004031 3					-30 | +115 Klübersynth UH1 6-150			
	-25 +80		CLP PG rPCF (PSS)	VG 220	-25 | +115 GearFluid Poly 220 E1 SEW07004031 3								
	[4] -25 +40		CLP PG NSF H1 (PSS)	VG 220	-25 | +95 GearOil Poly 220 H1 E1 SEW07004031 3					-25 | +95 Klübersynth UH1 6-220			
	-20 +80			VG 460	-20 | +115 GearOil Poly 460 H1 E1 SEW07004031 3					-20 | +115 Klübersynth UH1 6-460			
	-30 +30			VG 150	-30 | +80 GearOil Poly 150 H1 E1 SEW07004031 3					-30 | +80 Klübersynth UH1 6 - 50			

[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🖹 200).

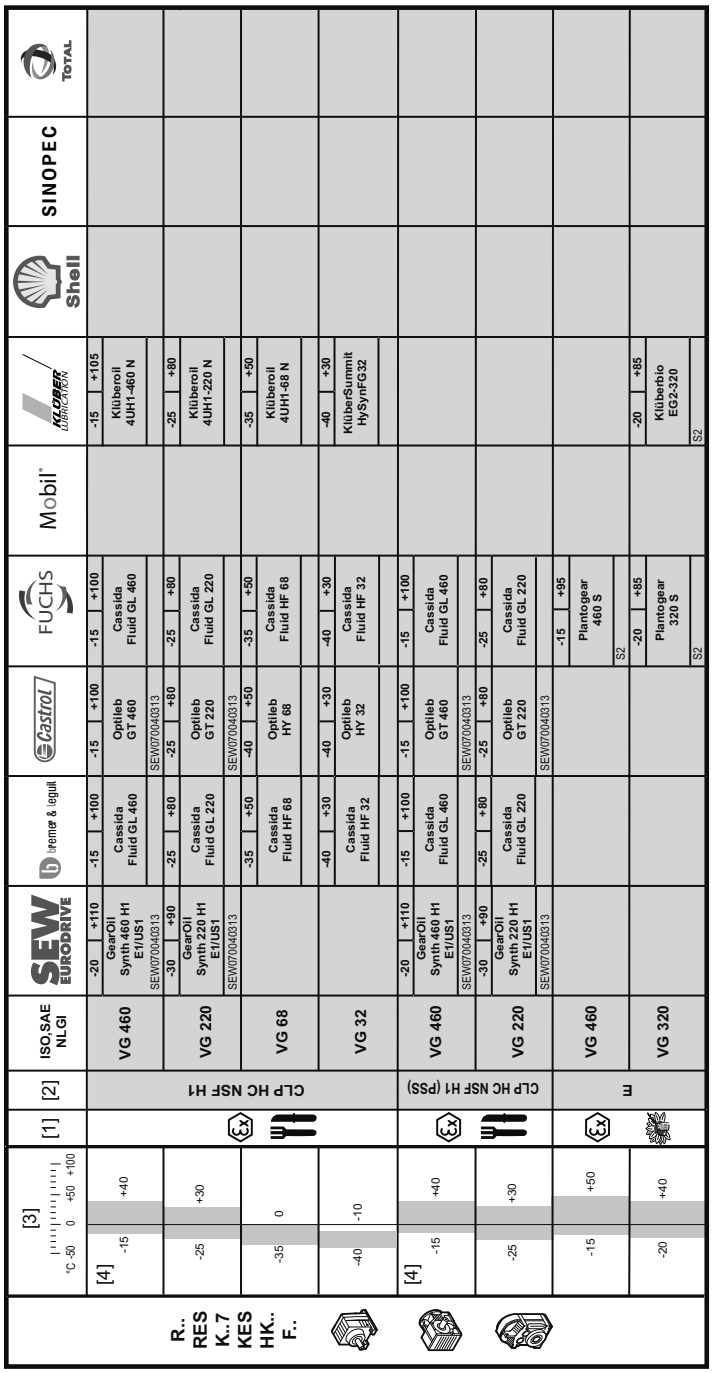


[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).



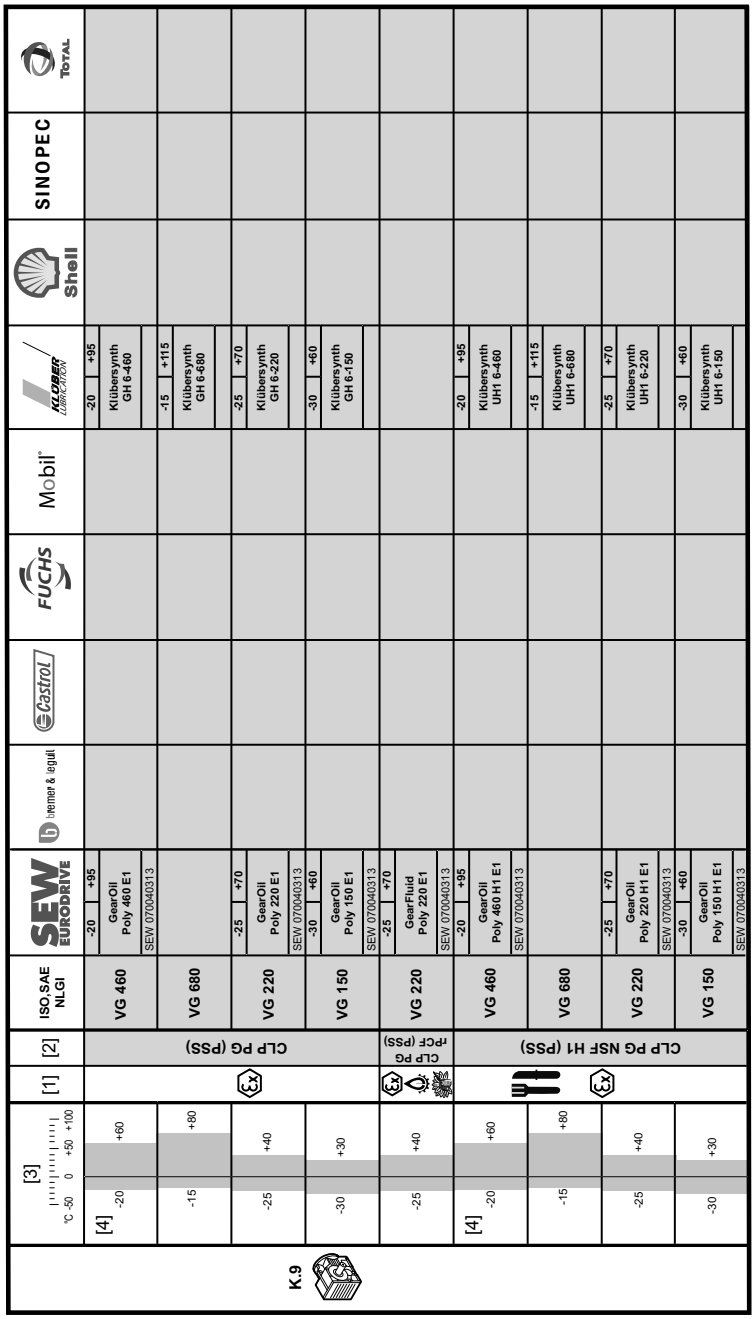
[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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Lubricant table for K..9 gear units

The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).



K.9	[3] °C -50 0 +50 +100	[1]	[2]	ISO, SAE NLGI	SEW EURODRIVE	Lubrication	Castrol	FUCHS	Mobil	KLÜBER LUBRICATION	Shell	SINOPEC	TOTAL
[4]	-20 +60	Ex	CLP PG (PSS)	VG 460	-20 | +95 GearOil Poly 460 E1 SEW 070040313					-20 | +95 Klübersynth GH 6-460			
	-15 +80			VG 680						-15 | +115 Klübersynth GH 6-680			
	-25 +40			VG 220	-25 | +70 GearOil Poly 220 E1 SEW 070040313					-25 | +70 Klübersynth GH 6-220			
	-30 +30			VG 150	-30 | +60 GearOil Poly 150 E1 SEW 070040313					-30 | +60 Klübersynth GH 6-150			
	-25 +40	Ex	CLP PG rPCF (PSS)	VG 220	-25 | +70 GearFluid Poly 220 E1 SEW 070040313								
[4]	-20 +60	Ex	CLP PG NSF H1 (PSS)	VG 460	-20 | +95 GearOil Poly 460 H1 E1 SEW 070040313					-20 | +95 Klübersynth UH1 6-460			
	-15 +80			VG 680						-15 | +115 Klübersynth UH1 6-680			
	-25 +40			VG 220	-25 | +70 GearOil Poly 220 H1 E1 SEW 070040313					-25 | +70 Klübersynth UH1 6-220			
	-30 +30			VG 150	-30 | +60 GearOil Poly 150 H1 E1 SEW 070040313					-30 | +60 Klübersynth UH1 6-150			

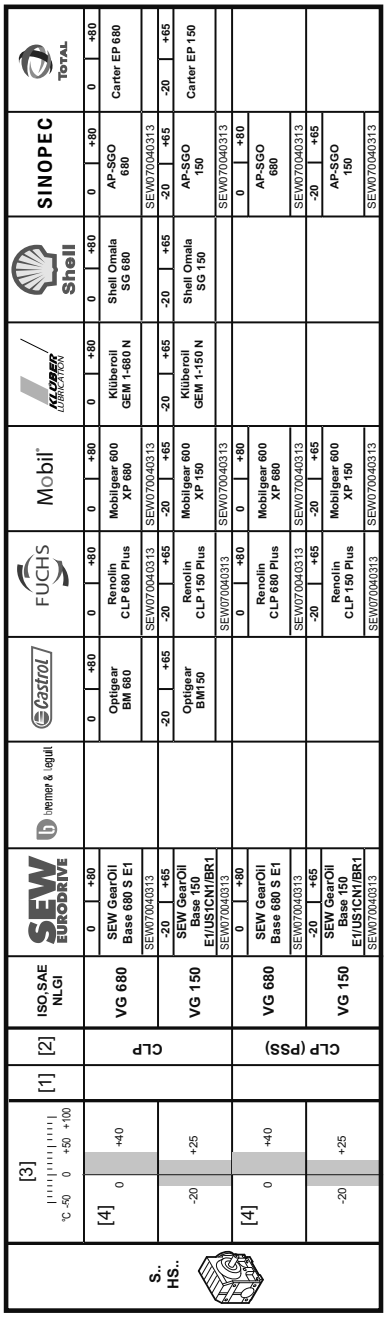
[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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Lubricant table for S.. gear units

The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).

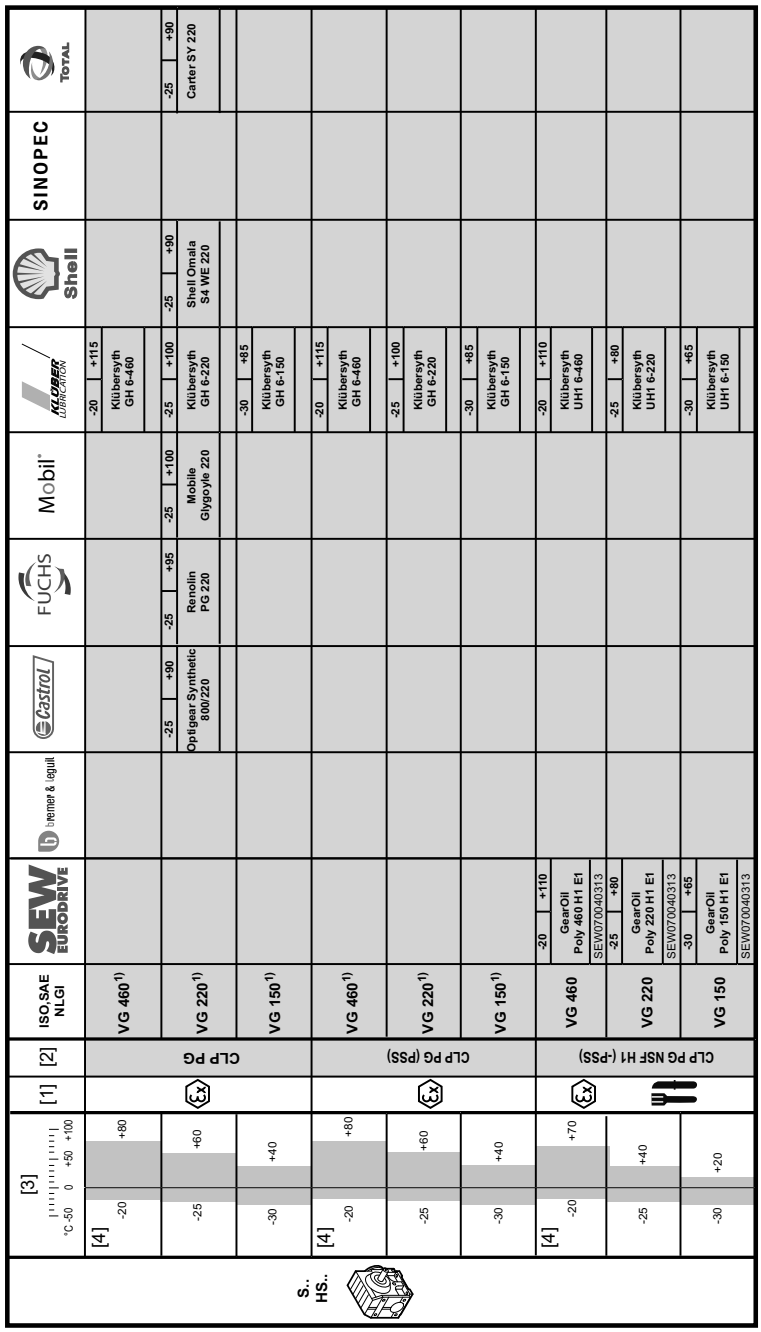


[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).

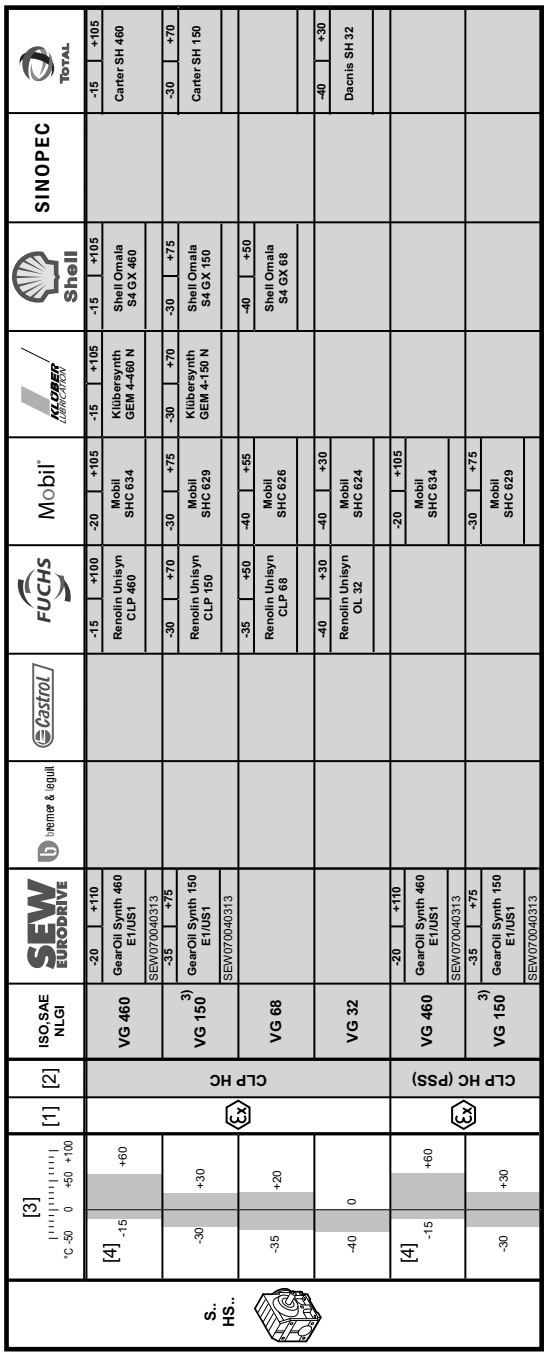


[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).



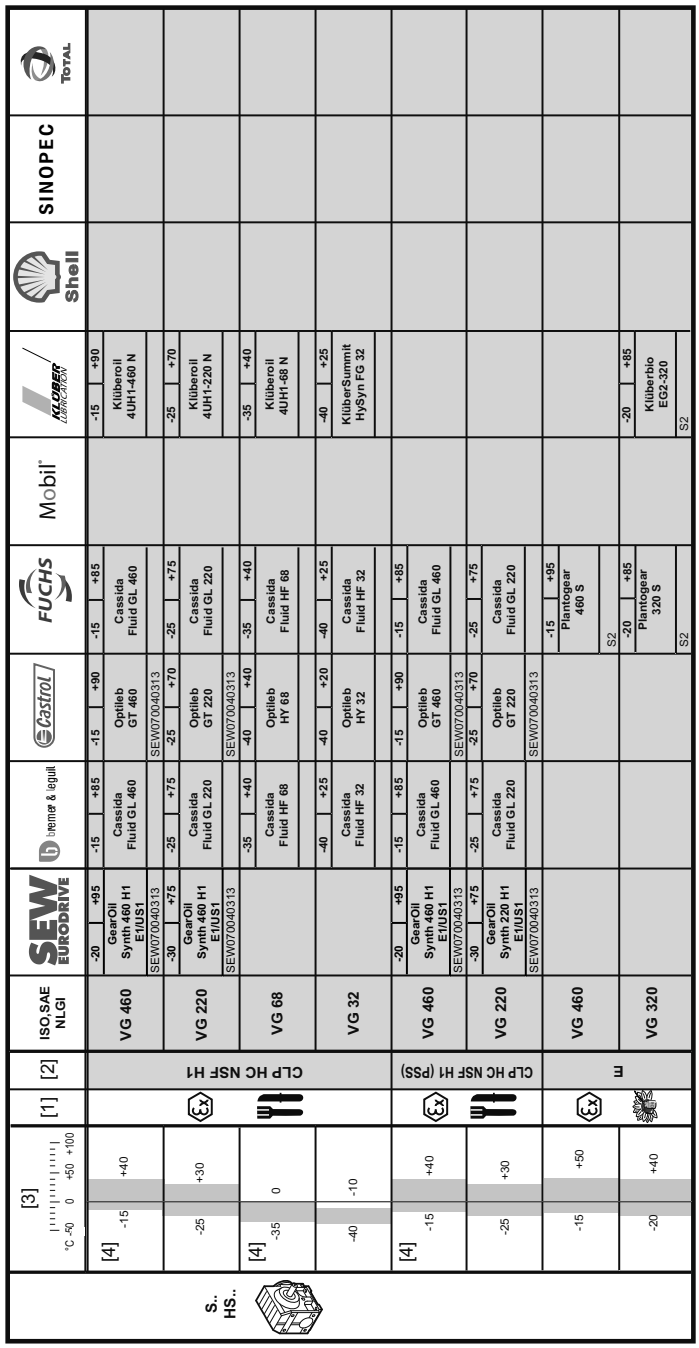
S.. HS..	[3] °C -50 0 +50 +100	[1]	[2]	ISO,SAE NLGI	SEW EURODRIVE	Castrol	FUCHS	Mobil	Klüber Lubrication	Shell	SINOPEC	TOTAL
	[4] -15 +60	Ex	CLP HC	VG 460	-20 | +110 GearOil Synth 460 E1/US1 SEW07004031 3		-15 | +100 Renolin Unisyn CLP 460	-20 | +105 Mobil SHC 634	-15 | +105 Klübersynth GEM 4-460 N	-15 | +105 Shell Omala S4 GX 460		-15 | +105 Carter SH 460
	-30 +30			VG 150 3)	-35 | +75 GearOil Synth 150 E1/US1 SEW07004031 3		-30 | +70 Renolin Unisyn CLP 150	-30 | +75 Mobil SHC 629	-30 | +70 Klübersynth GEM 4-150 N	-30 | +75 Shell Omala S4 GX 150		-30 | +70 Carter SH 150
	-35 +20			VG 68			-35 | +50 Renolin Unisyn CLP 68	-40 | +55 Mobil SHC 626		-40 | +50 Shell Omala S4 GX 68		
	-40 0			VG 32			-40 | +30 Renolin Unisyn OL 32	-40 | +30 Mobil SHC 624				-40 | +30 Dacnis SH 32
	[4] -15 +60	Ex	CLP HC (PSS)	VG 460	-20 | +110 GearOil Synth 460 E1/US1 SEW07004031 3			-20 | +105 Mobil SHC 634				
	-30 +30			VG 150 3)	-35 | +75 GearOil Synth 150 E1/US1 SEW07004031 3			-30 | +75 Mobil SHC 629				

[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).

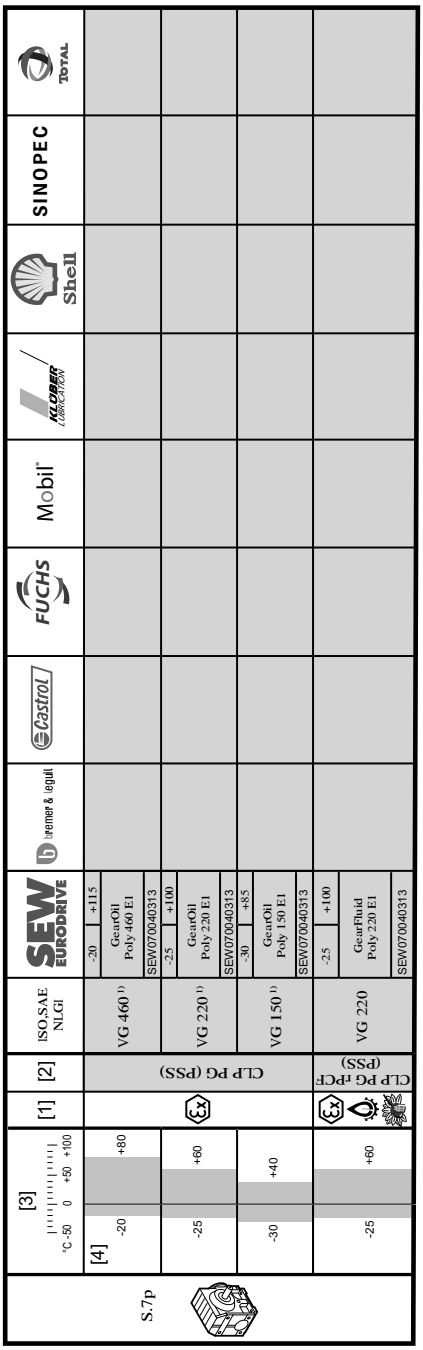


S.. HS..	[3] °C -50 0 +50 +100	[1]	[2]	ISO,SAE NLGI	SEW EURODRIVE	Lubrimor & Lagutt	Castrol	FUCHS	Mobil	KLÜBER LUBRICATION	Shell	SINOPEC	TOTAL
	[4] -15 +40	Ex	CLP HC NSF H1	VG 460	-20 +95 GearOil Synth 460 H1 E1/US1	-15 +85 Cassida Fluid GL 460	-15 +90 Optileb GT 460	-15 +85 Cassida Fluid GL 460		-15 +90 Klüberoil 4UH1-460 N			
					SEW07004031 3		SEW07004031 3						
	-25 +30			VG 220	-30 +75 GearOil Synth 460 H1 E1/US1	-25 +75 Cassida Fluid GL 220	-25 +70 Optileb GT 220	-25 +75 Cassida Fluid GL 220		-25 +70 Klüberoil 4UH1-220 N			
					SEW07004031 3		SEW07004031 3						
	[4] -35 0			VG 68		-35 +40 Cassida Fluid HF 68	-40 +40 Optileb HY 68	-35 +40 Cassida Fluid HF 68		-35 +40 Klüberoil 4UH1-68 N			
	-40 -10			VG 32		-40 +25 Cassida Fluid HF 32	-40 +20 Optileb HY 32	-40 +25 Cassida Fluid HF 32		-40 +25 KluberSummit HySyn FG 32			
	[4] -15 +40	Ex	CLP HC NSF H1 (PSS)	VG 460	-20 +95 GearOil Synth 460 H1 E1/US1	-15 +85 Cassida Fluid GL 460	-15 +90 Optileb GT 460	-15 +85 Cassida Fluid GL 460					
					SEW07004031 3		SEW07004031 3						
	-25 +30			VG 220	-30 +75 GearOil Synth 220 H1 E1/US1	-25 +75 Cassida Fluid GL 220	-25 +70 Optileb GT 220	-25 +75 Cassida Fluid GL 220					
					SEW07004031 3		SEW07004031 3						
	-15 +50	Ex	E	VG 460				-15 +95 Plantogear 460 S					
								S2					
	-20 +40			VG 320				-20 +85 Plantogear 320 S		-20 +85 Klüberbio EG2-320			
								S2		S2			

[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).



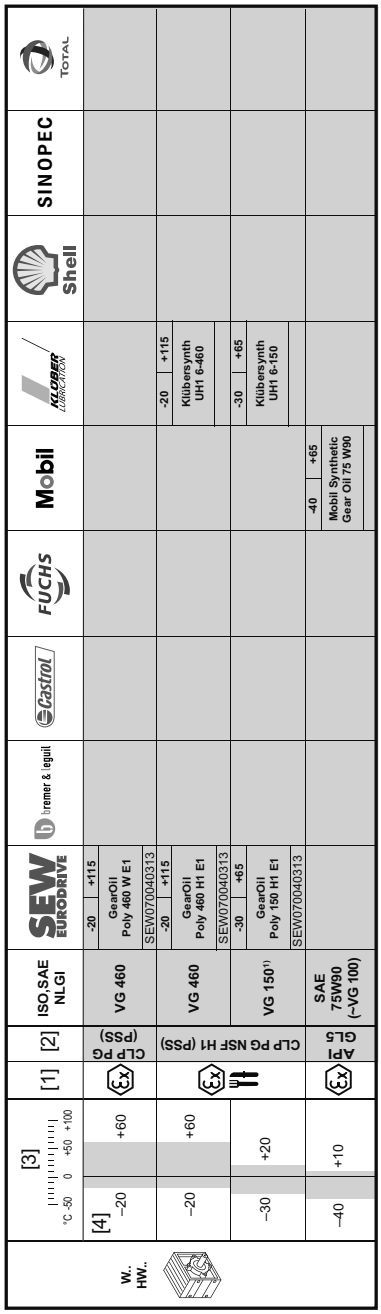
[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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Lubricant table for W.. gear units

The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🗎 200).



W.. HW..	[3] °C -50 0 +50 +100	[4]	[1]	[2]	ISO SAE NLGI	SEW EURODRIVE	bremer & leguil	Castrol	FUCHS	Mobil	KLÜBER LUBRICATION	Shell	SINOPEC	TOTAL
		–20		CLP PG (PSS)	VG 460	-20 | +115 GearOil Poly 460 W E1 SEW07004031 3								
		–20		CLP PG NSF H1 (PSS)	VG 460	-20 | +115 GearOil Poly 460 H1 E1 SEW07004031 3					-20 | +115 Klübersynth UH1 6-460			
		–30			VG 150 [1]	-30 | +65 GearOil Poly 150 H1 E1 SEW07004031 3					-30 | +65 Klübersynth UH1 6-150			
		–40		API GL5	SAE 75W90 (~VG 100)					-40 | +65 Mobil Synthetic Gear Oil 75 W90				

[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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The lubricant table is valid on the day this document is published. See **www.sew-eurodrive.de/lubricants** for the latest tables.

Observe the thermal limit of the oil seal material, see chapter "Lubricant compatibility with oil seals" (→ 🖹 200).

	[3] °C -50 0 +50 +100	[1]	[2]	ISO,SAE NLGI	SEW EURODRIVE	Lubrication Lubmaster	Mobil	FUCHS	Castrol	Lubmaster	SINOPEC	Shell	TOTAL
W.9	[4] –25		CLP PG (PSS)	VG 220	-25 | +100 GearOil Poly 220 E1 SEW07004031	-25 | +100 Klübersynth GH 6-220							
	–20			VG 460	-20 | +115 GearOil Poly 460 E1 SEW07004031	-20 | +115 Klübersynth GH 6-460							
	–30			VG 150	-30 | +85 GearOil Poly 150 H1 E1 SEW07004031	-30 | +85 Klübersynth GH 6-150							
	–25		CLP PG rPCF (PSS)	VG 220	-25 | +100 GearFluid Poly 220 E1 SEW07004031								
	[4] –25		CLP PG NSF H1 (PSS)	VG 220	-25 | +80 GearOil Poly 220 H1 E1 SEW07004031	-25 | +80 Klübersynth UH 6-220							
	–20			VG 460	-20 | +110 GearOil Poly 460 H1 E1 SEW07004031	-20 | +110 Klübersynth UH 6-460							
	–30			VG 150	-30 | +65 GearOil Poly 150 H1 E1 SEW07004031	-30 | +65 Klübersynth UH 6-150							

[1] Note on special approvals
[2] Oil type
[3] Ambient temperature range
[4] Standard

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8.2.3 Lubricant fill quantities

The following tables show guide values for lubricant fill quantities in relation to mounting positions M1 to M6.

The specified fill quantities are guide values. The exact values vary depending on the number of gear stages and the gear ratio. Always check the oil level plug during filling for the exact oil quantity.

Unless a special arrangement is made, SEW-EURODRIVE supplies the drives with a lubricant fill adapted for the specific mounting position. The mounting position (see chapter "Designation of the mounting positions" (→ 🗎 155)) must therefore be specified in the drive order.

When the mounting position is changed, the lubricant fill quantity must be adapted accordingly (see the following chapters). Do not change the mounting position without prior consultation with SEW-EURODRIVE, otherwise the rights to claim under limited warranty become void.

Helical (R) gear units

R.., R..F

Gear unit	Fill quantity in liters					
	M1[1)]	M2[1)]	M3[1)]	M4	M5	M6
R07	0.12	0.20				
R17	0.25	0.55	0.35	0.55	0.35	0.40
R27	0.25/0.40	0.70	0.50	0.70	0.50	
R37	0.30/0.95	0.85	0.95	1.05	0.75	0.95
R47	0.70/1.50	1.60	1.50	1.65	1.50	
R57	0.80/1.70	1.90	1.70	2.10	1.70	
R67	1.10/2.30	2.40	2.80	2.90	1.80	2.00
R77	1.20/3.00	3.30	3.60	3.80	2.50	3.40
R87	2.30/6.0	6.5/8.1	7.4/7.2	7.4	6.4	6.6
R97	4.60/9.8	11.7		13.4	11.3	11.7
R107	6.0/13.7	16.3	16.9	19.2	13.2	15.9
R127	6.4/17	18.3	18.2	22.0	16.8	17.9
R137	10.0/25.0	28.0	29.5	31.5	25.0	
R147	15.4/40.0	46.5	48.0	52.0	39.5	41.0
R167	27.0/70.0	82.0	78.0	88.0	66.0	69.0

1) With compound gear units, the large gear unit must be filled with the larger oil quantity.

RF.., RM.., RZ..

Gear unit	Fill quantity in liters					
	M1[1)]	M2[1)]	M3	M4	M5	M6
RF07	0.12	0.20				
RF17	0.25	0.55	0.35	0.55	0.35	0.40
RF27	0.25/0.40	0.70	0.50	0.70	0.50	
RF37	0.35/0.95	0.90	0.95	1.05	0.75	0.95
RF47	0.65/1.50	1.60	1.50	1.65	1.50	
RF57	0.80/1.70	1.80	1.70	2.00	1.70	
RF67	1.20/2.50	2.50/3.2	2.70	2.80	1.90	2.10
RF77	1.20/2.60	3.10/4.0	3.30	3.60	2.40	3.00
RF87	2.40/6.0	6.5/8.2	7.3	7.4	6.4	6.5
RF97	5.1/10.2	11.9	11.2	14.0	11.2	11.8
RF107	6.3/14.9	15.9	17.0	19.2	13.1	15.9
RF127	6.6/16.0	18.3	18.2	21.4	15.9	17.0
RF137	9.5/25.0	27.0	29.0	32.5	25.0	

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Gear unit	Fill quantity in liters					
	M1[1)]	M2[1)]	M3	M4	M5	M6
RF147	16.4/42.0	47.0	48.0	52.0	42.0	
RF167	26.0/70.0	82.0	78.0	88.0	65.0	71.0

1) With compound gear units, the large gear unit must be filled with the larger oil quantity.

RX..

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
RX57	0.60	0.80	1.30		0.90	
RX67	0.80		1.70	1.40	1.10	
RX77	1.10	1.50	2.60	2.70	1.60	
RX87	1.70	2.50	4.80		2.90	
RX97	2.10	3.40	7.4	7.0	4.80	
RX107	3.90	5.6	11.6	11.9	7.7	

RXF..

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
RXF57	0.50	0.80	1.10		0.70	
RXF67	0.70	0.80	1.50	1.40	1.00	
RXF77	0.90	1.30	2.40	2.00	1.60	
RXF87	1.60	1.95	4.90	3.95	2.90	
RXF97	2.10	3.70	7.1	6.3	4.80	
RXF107	3.10	5.7	11.2	9.3	7.2	

Parallel shaft helical (F) gear units

F.., FA..B, FH..B, FV..B

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
F..27	0.60	0.80	0.65	0.70	0.60	
F..37	0.95	1.25	0.70	1.25	1.00	1.10
F..47	1.50	1.80	1.10	1.90	1.50	1.70
F..57	2.25	3.15	1.65	3.15	2.40	2.50
F..67	2.70	3.80	1.90	3.80	2.90	3.20
F..77	5.90	7.30	4.30	8.00	6.00	6.30
F..87	11.0	13.1	7.70	14.0	10.9	11.1
F..97	18.8	22.7	12.6	25.5	18.6	20.2
F..107	24.5	32.0	19.5	37.5	27.0	
F..127	40.5	54.5	34.0	61.0	46.3	47.0
F..157	74.0	106.5	63.0	110.0	88.5	80.5

FF..

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
FF27	0.60	0.80	0.65	0.70	0.60	
FF37	1.00	1.25	0.70	1.30	1.00	1.10
FF47	1.60	1.85	1.10	1.90	1.50	1.70
FF57	2.30	3.05	1.70	3.10	2.30	2.40
FF67	2.70	3.80	1.90	3.80	2.90	3.20
FF77	5.90	7.30	4.30	8.10	6.00	6.30
FF87	11.0	13.3	7.80	14.3	11.1	11.3

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Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
FF97	19.3	22.7	12.6	25.9	19.0	20.7
FF107	25.5	32.0	19.5	38.5	18.6	28.0
FF127	41.5	55.5	34.0	63.0	45.0	49.0
FF157	77.0	107.5	64.0	111.0	89.5	81.5

FA.., FH.., FV.., FAF.., FAZ.., FHF.., FZ.., FHZ.., FVF.., FVZ.., FT.., FM.., FAM..

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
F..27	0.60	0.80	0.65	0.70	0.60	
F..37	0.95	1.25	0.70	1.25	1.00	1.10
F..47	1.50	1.80	1.10	1.90	1.50	1.70
F..57	2.40	3.10	1.70	3.15	2.40	2.50
F..67	2.70	3.80	1.90	3.80	2.90	3.20
F..77	5.90	7.30	4.30	8.00	6.00	6.30
F..87	11.0	13.1	7.70	14.0	10.9	11.1
F..97	18.8	22.7	12.6	25.5	18.6	20.2
F..107	24.5	32.0	19.5	37.5	27.0	
F..127	39.0	54.5	34.0	61.0	45.0	46.5
F..157	73.0	105.5	62.0	109.0	87.5	79.5

Helical-bevel (K) gear units

K..19 and K..29 gear units have a universal design and are filled with the same quantity of oil regardless of the mounting position, with the exception of M4.

K.., KA..B, KH..B, KV..B

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
K..19	0.40			0.45	0.40	
K..29	0.70			0.85	0.70	
K..37	0.50	1.00		1.25	0.95	
K..39	0.86	1.65	1.55	2.10	1.55	1.30
K..47	0.80	1.30	1.50	2.00	1.60	
K..49	1.65	3.35	2.80	4.20	3.15	2.75
K..57	1.10	2.20		2.80	2.30	2.10
K..67	1.10	2.40	2.60	3.45	2.60	
K..77	2.20	4.10	4.40	5.80	4.20	4.40
K..87	3.70	8.20	8.90	10.90	8.20	
K..97	7.0	14.0	15.70	20.0	15.70	15.50
K..107	10.0	21.0	25.50	33.50	24.0	
K..127	19.0	41.50	44.0	54.0	40.0	41.0
K..157	31.0	65.0	68.0	90.0	62.0	63.0
K..167	33.0	97.0	109.0	127.0	89.0	86.0
K..187	53.0	156.0	174.0	207.0	150.0	147.0

KF..

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
KF19	0.40			0.45	0.40	
KF29	0.70			0.85	0.70	
KF37	0.50	1.10		1.40	1.00	
KF39	0.86	1.65	1.55	2.10	1.55	1.30

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Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
KF47	0.80	1.30	1.70	2.20	1.60	
KF49	1.65	3.35	2.80	4.20	3.15	2.75
KF57	1.20	2.20	2.40	3.15	2.50	2.30
KF67	1.10	2.40	2.80	3.70	2.70	
KF77	2.10	4.10	4.40	5.90	4.50	
KF87	3.70	8.30	9.2	11.90	8.60	8.50
KF97	7.0	14.70	17.30	21.50	15.70	16.50
KF107	10.0	21.90	26.0	35.10	25.40	25.30
KF127	19.0	41.50	46.0	55.0	41.0	
KF157	31.0	66.0	69.0	92.0	63.0	

KA.., KH.., KV.., KAF.., KHF.., KVF.., KZ.., KAZ.., KHZ.., KVZ.., KT.., KM.., KAM..

Gear unit	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
K..19	0.40			0.45	0.40	
K..29	0.70			0.85	0.70	
K..37	0.50	1.00		1.30	1.00	
K..39	0.86	1.65	1.55	2.10	1.55	1.30
K..47	0.80	1.30	1.60	2.0	1.60	
K..49	1.65	3.35	2.80	4.20	3.15	2.75
K..57	1.20	2.20	2.40	3.15	2.70	2.40
K..67	1.10	2.40	2.70	3.70	2.60	
K..77	2.10	4.10	4.60	5.90	4.40	
K..87	3.70	8.40	9.0	11.10	8.2	
K..97	7.0	14.70	15.70	20.0	15.70	
K..107	10.0	20.80	24.5	32.4	24.5	24.3
K..127	19.0	41.50	43.0	52.0	40.0	
K..157	31.0	65.0	68.0	90.0	62.0	63.0
K..167	33.0	97.0	109.0	127.0	89.0	86.0
K..187	53.0	156.0	174.0	207.0	150.0	147.0

Helical-worm (S) gear units

S..

Gear unit	Fill quantity in liters					
	M1	M2	M3[1)]	M4	M5	M6
S37	0.25	0.40	0.50	0.55	0.40	
S47	0.35	0.80	0.70/0.90	1.00	0.80	
S57	0.50	1.20	1.00/1.20	1.35	1.30	
S67	1.00	2.00	2.20/3.10	3.10	2.60	2.60
S77	1.90	4.20	3.70/5.4	5.9	4.40	
S87	3.30	8.1	6.9/10.4	11.3	8.4	
S97	6.8	15.0	13.4/18.0	21.8	17.0	

1) With compound gear units, the large gear unit must be filled with the larger oil quantity.

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SF..

Gear unit	Fill quantity in liters										
	M1	M2	M3[1)]	M4 Output		M5 Output			M6 Output		
				A/B	A + B	A	B	A + B	A	B	A + B
SF37	0.25	0.40	0.50	0.55	0.6	0.4	0.4	0.4	0.4	0.4	0.4
SF47	0.40	0.90	0.90/1.05	1.10	1.15	1.0	0.9	1.0	0.9	1.0	1.0
SF57	0.50	1.20	1.00/1.50	1.50	1.55	1.4	1.4	1.4	1.4	1.4	1.4
SF67	1.00	2.20	2.30/3.00	3.20	3.5	2.7	2.6	2.7	2.6	2.7	2.7
SF77	1.90	4.10	3.90/5.8	6.5	7.2	4.9	4.6	4.9	4.6	4.9	4.9
SF87	3.80	8.0	7.1/10.1	12.0	13.2	9.1	8.2	9.1	8.2	9.1	9.1
SF97	7.4	15.0	13.8/18.8	23.1	25.2	18.0	17.0	18.0	17.0	18.0	18.0

1) With compound gear units, the large gear unit must be filled with the larger oil quantity.

SA.., SH.., SAF.., SHZ.., SAZ.., SHF.., ST..

Gear unit	Fill quantity in liters					
	M1	M2	M3[1)]	M4	M5	M6
S..37	0.25	0.40	0.50		0.40	
S..47	0.40	0.80	0.70/0.90	1.05	0.80	
S..57	0.50	1.10	1.00/1.50	1.45	1.20	
S..67	1.00	2.00	1.80/2.60	2.90	2.50	
S..77	1.80	3.90	3.60/5.0	5.8	4.50	
S..87	3.80	7.4	6.0/8.7	10.8	8.0	
S..97	7.0	14.0	11.4/16.0	21.0	15.7	

1) With compound gear units, the large gear unit must be filled with the larger oil quantity.

SPIROPLAN® (W) gear units

SPIROPLAN® gear units W..10 to W..30 have a universal design and are filled with the same quantity of oil regardless of the mounting position.

The oil fill quantity of SPIROPLAN® gear units W..37 and W..47 in mounting position M4 is different from that of the other mounting positions.

W.., WA..B, WH..B

Gear unit	Fill quantity in liters						
	M1	M2	M3	M4 2	M4 3	M5	M6
W..10	0.16						
W..20	0.24						
W..30	0.40						
W..37	0.50			0.70		0.50	
W..47	0.90			1.40		0.90	

WF..

Gear unit	Fill quantity in liters						
	M1	M2	M3	M4 2	M4 3	M5	M6
W..10	0.16						
W..19	0.34			0.57	0.51	0.515	
W..20	0.24						
W..29	0.54			0.93	0.78	0.72	
W..30	0.40						
W..37	0.50			0.70		0.50	

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Gear unit	Fill quantity in liters						
	M1	M2	M3	M4		M5	M6
				2	3		
W..39	0.85			1.5	1.35	1.25	
W..47	0.90			1.55		0.90	
W..49	1.39			2.41	2.19	2.15	
W..59	2.0			3.49	3.2	3.0	

WA.., WAF..,WH.., WT.., WHF..

Gear unit	Fill quantity in liters						
	M1	M2	M3	M4		M5	M6
				2	3		
W..10	0.16						
W..19	0.34			0.57	0.51	0.515	
W..20	0.24						
W..29	0.54			0.93	0.78	0.72	
W..30	0.40						
W..37	0.50			0.70		0.50	
W..39	0.85			1.5	1.35	1.25	
W..47	0.80			1.40		0.80	
W..49	1.39			2.41	2.19	2.15	
W..59	2.0			3.49	3.2	3.0	

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